

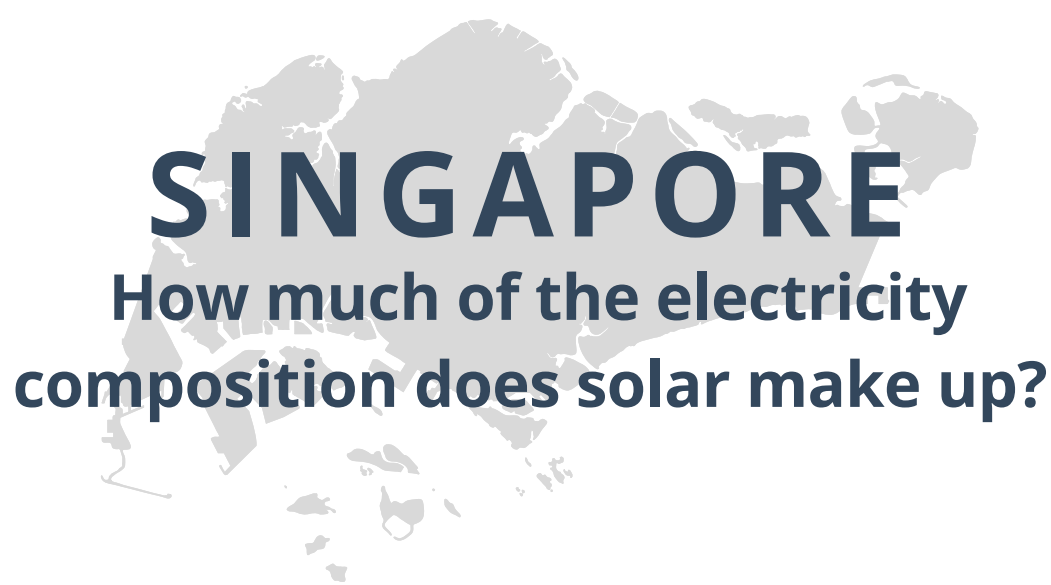


SOLAR ENERGY

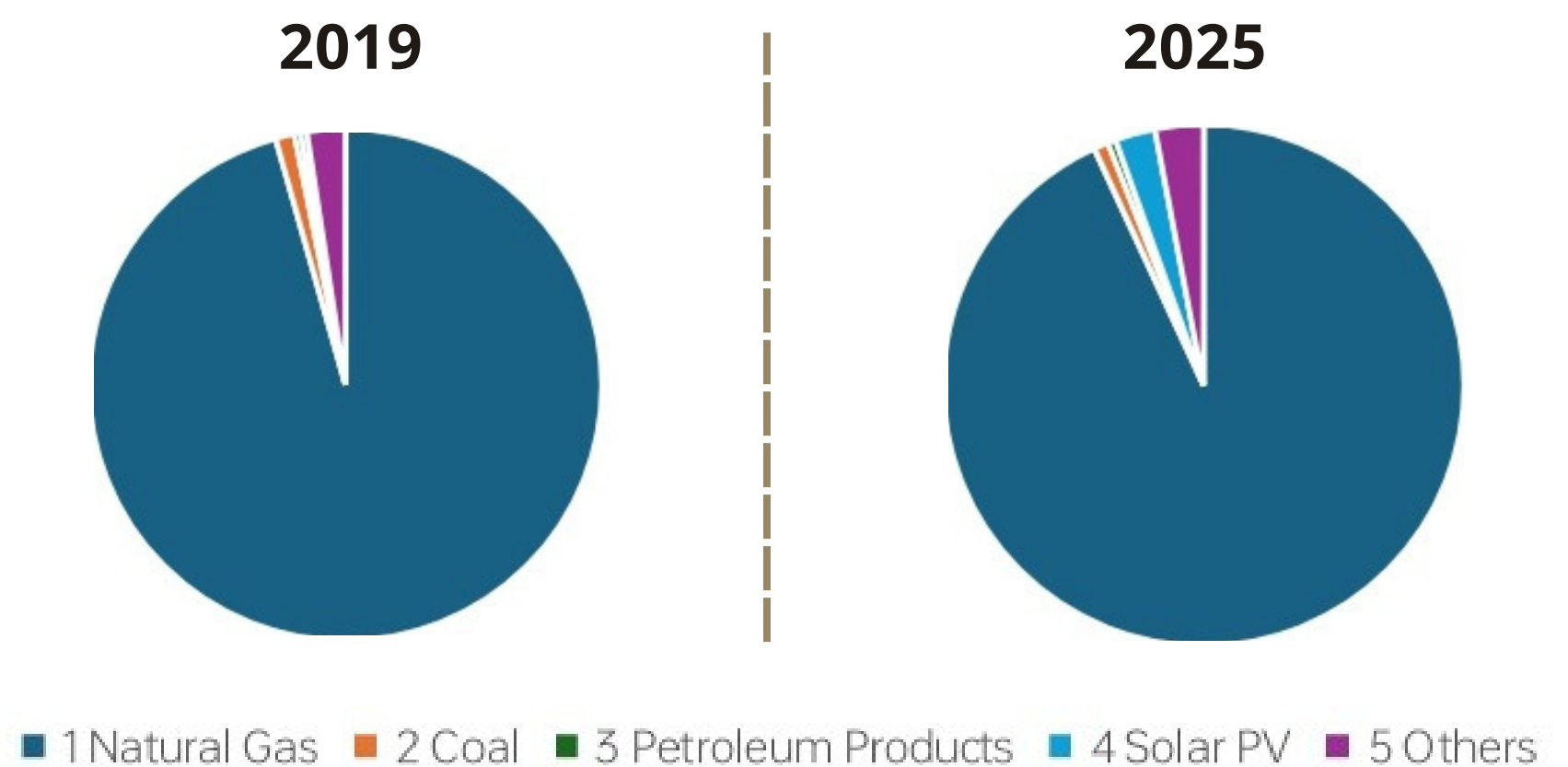
AN INTRODUCTION TO SOLAR ENERGY IN SINGAPORE

WHAT IS SOLAR ENERGY? ★

Solar energy is the radiant heat and light from the sun that is harnessed and converted into useful forms of energy, such as electricity or heat.



ELECTRICITY COMPOSITION MIX ★



Comparing the data from 2019 with the data from 1H of 2025, we can conclude that energy generation/ composition of Solar PV increased from 0.4% → 2.5% — indicating a 425% growth rate.

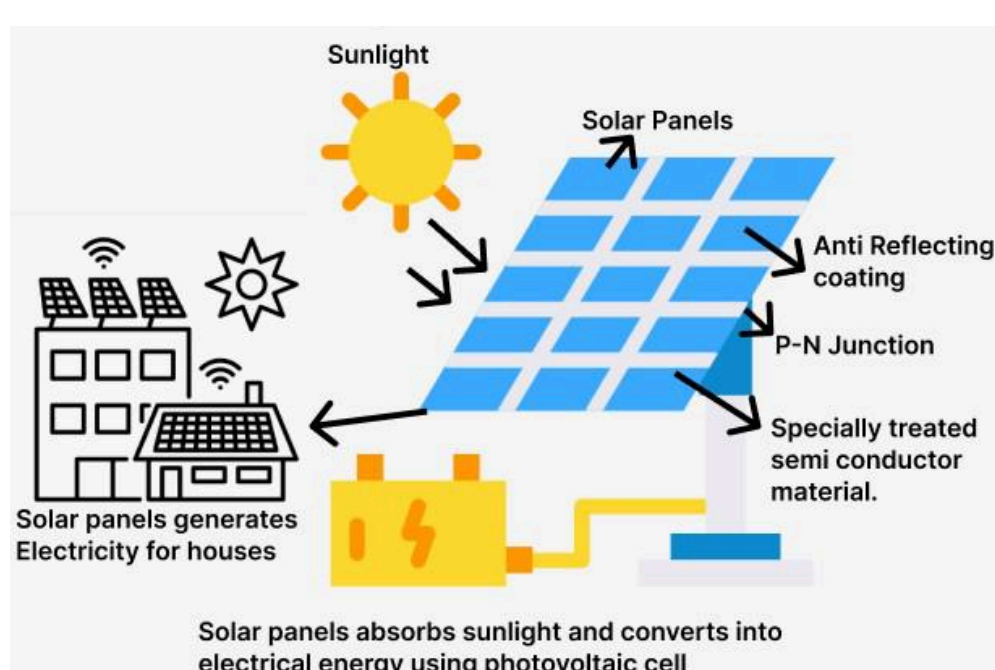
WHY SOLAR? ★

Wind	• Singapore's winds are not strong enough to provide sufficient energy • Turbine blades may kill birds
Hydroelectric	• Singapore lacks huge/powerful rivers • Dams built for this purpose can negatively affect wildlife
Nuclear	• Singapore has a small landmass, which means that if a nuclear disaster occurs, the whole of Singapore might be implicated • Government is still looking into the safety, public opinion, as well as other aspects of it and
Solar	• Singapore is near the equator and thus has plentiful sunlight • Solar panels can be placed on roofs, helping with space conservation in land-scarce Singapore

THE FUTURE OF SOLAR ENERGY

Solar cells convert light energy into electric energy. However, traditionally used solar cells face wear and tear, and eventually, break down to release toxic chemicals. To make matters worse, the silicon used to make them cannot be used again. As such, researchers are turning to organic solar cells, which should contribute to less toxic waste, but the efficiency of those solar cells are still being fine-tuned.

MECHANISM OF SOLAR CELL ★



The photons are absorbed by the cell's semi-conductive material

The electrons in the solar cell's anode are excited

Sunlight contains photons, which hit the solar cells

The electrons flow towards the cathode, generating electricity

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